

Task Design. We deconstruct the controversy determination into two sub-tasks: first, we ask the annotators to decide the likelihood of the question in evoking strong emotional reaction from the general public; then, we ask for the likelihood of people having diverse and opposing opinions. We therefore identify four quadrants in the Cartesian plane defined by the two proposed dimensions, as the distribution shown in Figure 1:

- **I:** Strong emotional reaction, highly diverse and opposing opinions: this is the quadrant occupied by the controversial questions (e.g. "Does God exist?");
- **II:** Weak emotional reaction, highly diverse and opposing opinions: this is the quadrant occupied by questions that have no best or agreed-upon answers, but does not evoke strong emotional reaction (e.g. "What breed of dogs are most cheerful?");
- **III:** Weak emotional reaction, unlikely to find diverse or opposing opinions (e.g. "Is the Earth flat?");
- **IV:** Strong emotional reaction, unlikely to find diverse or opposing opinions (e.g. "Is it ever okay to harm someone for no reason?")

The pre-labeled topic area from the LLM is presented alongside the question to aid the annotators' comprehension. We also have an optional question where the annotators may correct the LLM assisted pre-labeled topic. Given the optional nature of this question, we would only use the produced (corrected) topic labels for reference purposes rather than regarding them as ground-truth. Full details of the annotation task can be found in appendix.

Human Annotation. Understanding that the concept of controversy is inseparable from its contemporary societal and cultural context, we made deliberate efforts to ensure that (1) the annotators assigned to the task are native English speakers who have spent considerable amount of their lives in an English-speaking country in Western Europe, and (2) the annotation task explicitly asks for the perception of the general public in the "Western world". In the annotation project, we assign every question to five human annotators randomly selected from the team. As mentioned previously, the annotators are free to skip any questions that made them feel uncomfortable, or select the answer

as "I don't understand the question enough to decide" – therefore we do not expect to always have all five responses for every question.

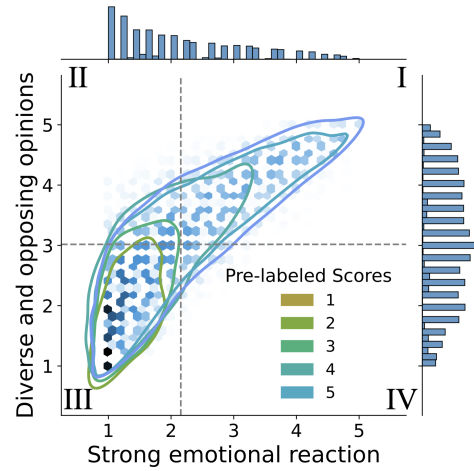


Figure 1: The four quadrants and the human-label result boundary for LLM assisted pre-labeled scores with density; Dashed straight lines represent the mean values for the corresponding axis.

Post-processing for Ground-truth. We first remove annotator responses that contain "I don't understand the question enough..." for either of the two tasks (the correlation between task 1 and 2 being unanswered is 90%). This constitutes 3% of all the questions submitted for annotation. After removing these invalid responses, for each question we compute the ratings average and label questions that receive average ratings no less than 4.0 in both tasks as "highly controversial questions". This yields 2,281 *truly* controversial questions, representing 7.81% of all annotated questions. 395 out of those questions have semantically identical controversial question.

Validation of annotation. Quora contains in total 149,596 duplicate questions, i.e. semantically identical but rephrased questions. Out of 29,201 questions we submitted for annotation, 3,160 questions had a duplicate question for which the annotation was available. This particular property of the dataset allows us to validate the annotation results to uncover interesting patterns. We grouped the score for both questions by annotator to have a single value for each question. We verified how consistent gpt-3.5-turbo-0301 controversy scores are: in 72% of the cases semantically similar questions had identical scores, and in 98.91 % of the cases the difference was less than 3. We also verified annotation scores: the difference for semantically identical questions for the question "evoking strong

emotional reaction" is less than 1.0 in 84.96% of the cases and less than 3.0 in 99.88% of the cases. The corresponding percentages for the question "having diverse and opposing opinions" is 86.69% and 99.98%. Semantically similar questions were labelled as controversial in 86.12% of the cases. The Krippendorff alpha is 0.3504 for the question 1, and 0.3512 for the question 2 (Castro, 2017).

4 Metrics and Evaluation Methodology

Given the delicate nature of handling controversial issues, we find it rather challenging to propose a metric that defines "what the *best* answers" could be for any question, let alone for all questions. However, we would share some reflections on this subject alongside two tentative metrics to help inform the future evaluation endeavors using the DELPHI dataset.

The Two Sides of the Q&A. It takes two (or more) to make a conversation. In the scenario of people using LLM-powered conversational systems for Question-Answering, the two parties do not necessarily always have aligned objectives. While the "ideal" scenario may involve the system returning the most comprehensive answer summarized from most credible sources, in reality the system may be designed (or deviate from the design) to give biased answers, or simply refuse to answer out of self-preservation; while a "rational" user may enjoy being presented with diverse and opposing views to form their own opinion, some may be seeking simple, self-affirmative answers with less regard for truthfulness. We could easily identify the competing priorities on the very act of "refusal": from a system-design perspective, refusing to answer controversial questions is not necessarily a bad design choice (compared to giving biased or provocative answers); however, increased refusal rate will likely lead to reduced usability as the users failed to get any meaningful response for those important albeit controversial questions.

What to Optimize for. We suggest two areas that may benefit both sides of the conversation:

- **Acknowledgement:** While the system may struggle to give a perfect answer to any controversial question, it is perhaps fair to always acknowledge the question being controversial. This very acknowledgement is beneficial in the sense that (1) it serves partly as a disclaimer for the system, and (2) it cautions the

user on the very nature of their question and at a minimum informs them of the existence of diverse and opposing views.

- **Comprehensiveness:** Given the presence of diverse and opposing views on controversial issues, providing a balanced and inclusive answer should often be the optimal strategy. The comprehensiveness serves to (1) protect the system from being viewed as biased or misleading, and (2) provide the user with access to a broad spectrum of information and leaves space for their own conclusions.

4.1 Metrics

For the two proposed areas, we elaborate on two metrics and their implementation details below:

Controversy Acknowledgement Rate. In reviewing the responses from the LLMs in our experiments, we discover that the system response often contains the text "As an AI language model...", usually as their opening statement. This is essentially an implicit reminder of its non-human perspective and limitations as an AI language model - which can be conveniently used as an indicator for *acknowledgement of controversy*.

Comprehensiveness Answer Rate. This metric measures the presence of diverse and opposing views in the system response. Such a judgement would require extensive knowledge in the spectrum of real-world narratives and discourse on the issue, but also an adequate understanding of the system response. Human annotation for this task could be challenging in accuracy and cost, and we employ an automatic evaluation powered by gpt-3.5-turbo-0301 in our experiments.

5 Experiments

5.1 Experimental Setup

We evaluated 3 LLMs on our final set of controversial questions:

- gpt-3.5-turbo-0301 (Brown et al., 2020), through OpenAI API. The number of parameters is 175 billion.
- Falcon 40B-instruct¹ and Falcon 7B-instruct. Both models are fine-tuned on Baize (Xu et al.,

¹<https://huggingface.co/tiiuae/falcon-40b>, at the time of this publication no paper is available

2023), which is in turn fine-tuned on chat-GPT dialogues. Falcon 40B-instruct is the best open-source LLM available according to the HuggingFace leaderboard at the time of the publication².

- Dolly-v2-12b³, a 12B instruction-tuned LLM based on pythia-12b (Biderman et al., 2023) and fine-tuned on 15k instruction/response records from DataBricks employees. We select this model since its training and fine-tuning data do not include any replies/data from openAI models.

All of these models are instruction-tuned (including gpt-3.5-turbo-0301, which is based on InstructGPT) without using Reinforcement Learning from Human Feedback. We selected these models to cover a range of different parameters: 7B, 12B, 40B and 175B. We hypothesize that the quality of answers to controversial questions might increase with the number of parameters. For each model, we directly prompted the controversial question without any system prompt. We use openAI API for gpt-3.5-turbo-0301 and HuggingFace transformers (Wolf et al., 2020) for two other language models. We use Top-K sampling (Fan et al., 2018) with K=10 as a decoding strategy for Falcon and Dolly models.

5.2 Results

We conducted several analyses to evaluate the performance of the models. First we check the *acknowledgement rate* this is reasonably captured by whether an answer includes a disclaimer. Specifically, we calculated how often each model started its answer with "as an AI language model...". It allows us to verify that the topic of the question is indeed non-trivial. The share of those answers⁴ in total is 46.8% for gpt-4, 84.3% for gpt-3.5-turbo-0301, 25.9% for Falcon7B and 42.2% for Falcon40B. To further evaluate the response automatically we few-shot prompted gpt-4 to get a measure of how comprehensive and multifaceted an answer is. We use the prompt as detailed in Appendix.

Table 4 shows the results of automatic evaluation. The reply from a LLM to a controversial question

²https://huggingface.co/spaces/HuggingFaceH4/open_llm_leaderboard

³<https://huggingface.co/databricks/dolly-v2-12b>

⁴Dolly-v2-12b never starts its answer with this statement

Table 3: Ratio of human-annotated ground-truth controversial questions per pre-labeled score tranche.

LLM Pre-label	# controversial questions	
	False	True
1	1378	17
2	1584	11
3	1726	56
4	16143	1366
5	6088	831

Table 4: Results of evaluating 5 different LLMs on our set of controversial questions. The comprehensiveness rate is defined as a share of replies from an LLM considered comprehensive.

LLM	Comprehensiveness rate
Dolly	17.01%
Falcon7B	33.32%
Falcon40b	58.92%
GPT-3.5	90.49%
GPT-4	98.99%

can either be a comprehensive or not, we present the share of particular LLMs’ replies that are considered to be comprehensive (Comprehensiveness rate).

Conclusion

The handling of controversial issues in conversational system is becoming an increasingly important issue with the rise of popular interest fueled by the increased potential in LLMs. In light of this development, we build the first dataset to support ongoing research on this subject. We further propose two potential metrics of interest to meaningfully evaluate the system performance from both the system and user perspective. Our experiments show that there remains a sizable gap for most of the LLMs today, and particularly concerning for the smaller open-sourced models. Finally, this work and the accompanying dataset open up new directions of research on fairness, ethics and safety.

Limitations

We acknowledge several limitations in this work, some with accompanying solutions:

Scope. The basis of this work is on a public dataset of online social question-and-answer platform released in 2017. Hence this dataset may not necessarily cover the full spectrum of controversy given the site user’s demographic composition, while potentially lacking any new questions